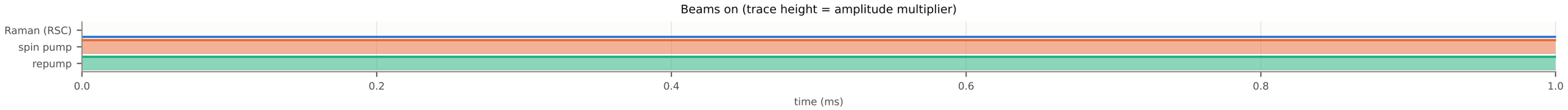
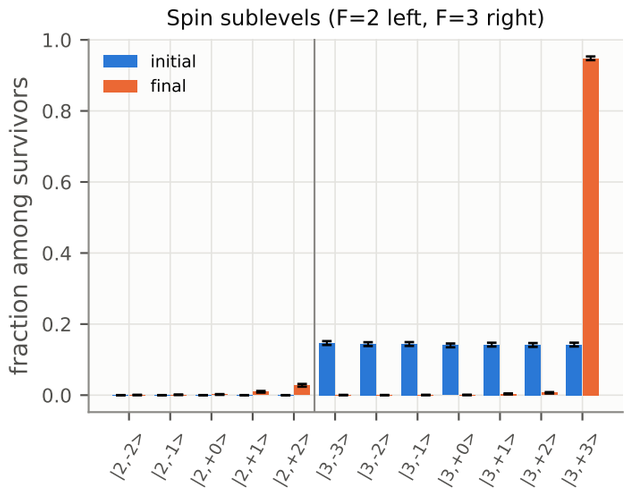
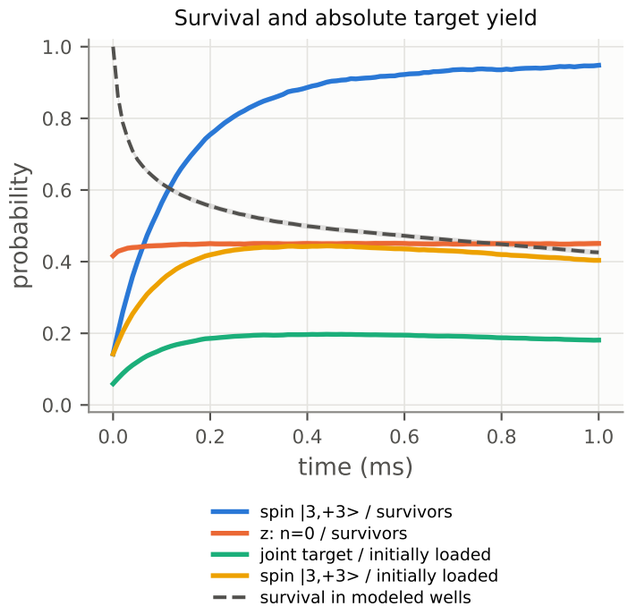
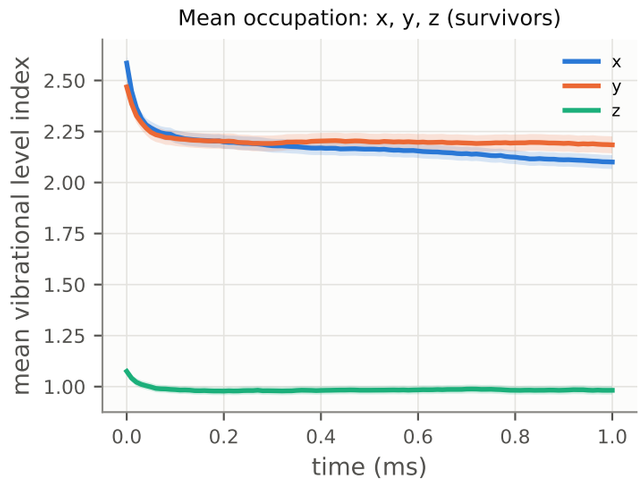
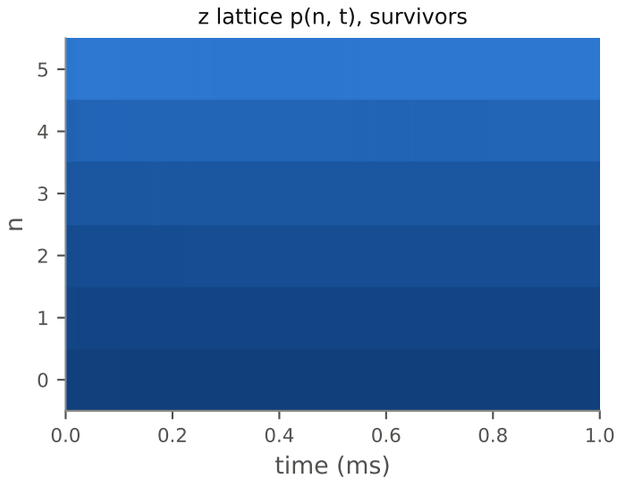
default
Model validity: WARNING

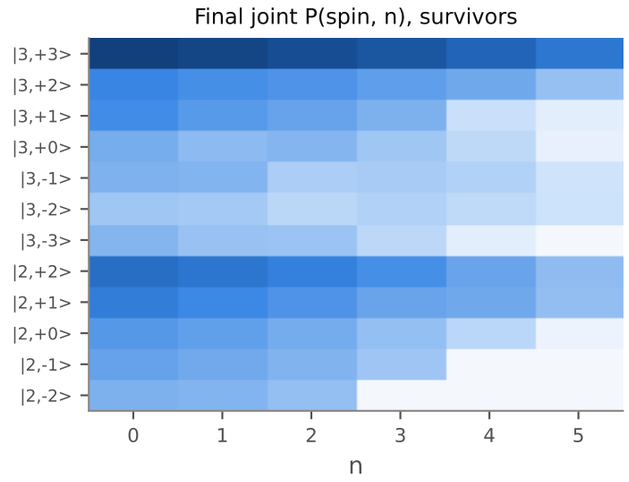


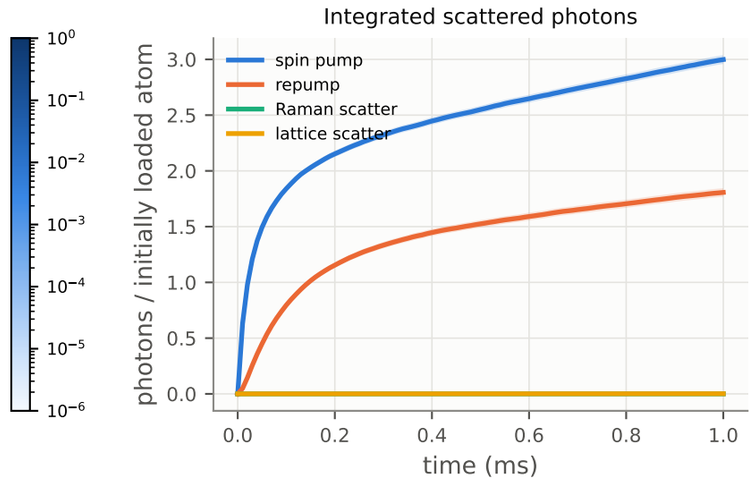


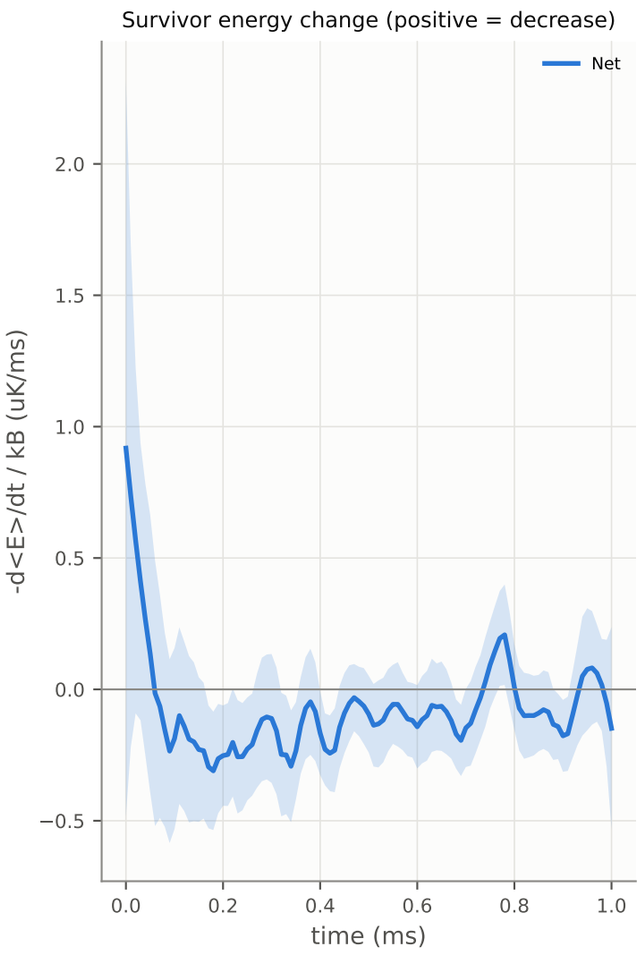













```
MODEL VALIDITY: WARNING
Solver: 3D coherent optical-pumping trajectories (Raman off).
Target: spin |3,+3>, local z ground state.
4 cloud sites x 4096 trajectories; trajectory seed 1729, spatial seed 0.
Means +/- one-sigma Monte Carlo standard errors.
Absolute probabilities use initially loaded bound atoms; conditional distributions use survivors.
Conditional errors include survival covariance. Thermal derivatives use trajectory batches with temporal covariance.
Temperature is an energy-matched proxy; net cooling includes loss selection (decomposition unavailable).

survival: 0.4259033 +/- 0.00373
loss: 0.5740967 +/- 0.00373
P_up_absolute: 0.4038086 +/- 0.0037
P_n0_absolute: 0.191975 +/- 0.00262
P_target_absolute: 0.1810516 +/- 0.00257
P_3d_ground_absolute: 0.01217575 +/- 0.000529
P_target_3d_absolute: 0.01118547 +/- 0.000513
photons_spin_pump: 2.99823 +/- 0.0203
photons_repump: 1.806763 +/- 0.0196
photons_lattice_scatter: 0.001342773 +/- 0.000286
photons_total: 4.806335 +/- 0.0367
nbar_x: 2.10043 +/- 0.0173
T_equiv_x_K: 6.086645e-06 +/- 1.05e-07
mean_excitation_energy_x_hz: 70276.6 +/- 586
nbar_y: 2.184516 +/- 0.0213
T_equiv_y_K: 6.733238e-06 +/- 1.37e-07
mean_excitation_energy_y_hz: 75903.75 +/- 749
nbar_z: 0.981912 +/- 0.0111
T_equiv_z_K: 4.382323e-06 +/- 6.83e-08
mean_excitation_energy_z_hz: 46887.98 +/- 557

Wall time: 2.92 s.

[OK] weak excitation: Maximum eliminated excited fraction across cloud sites = 0.004927.
[OK] ground coherences: Adjacent-spin angular splitting / conservative full photon-rate bound = 46.7; requires >> 1.
[OK] magnetic field: Ground, D1, and D2 hyperfine + Zeeman eigenstates and mixed optical dipoles; F,m labels follow the B=0 branches
adiabatically. Fine-structure mixing and diamagnetism omitted.
[OK] motional recoil: Coherent wavefunctions retained on x/y/z; the population motional-secular criterion does not apply to this
solver.
[OK] conservation: Maximum |survival + escape - 1| = 0.
[OK] positivity: Minimum final reduced-density eigenvalue = -6.65e-20.
[WARNING] Monte Carlo precision: 4096 trajectories per site; repeat with more trajectories and independent seeds. Rare unobserved
events may have zero empirical error.
[WARNING] spatial and angular quadrature: 4 frozen cloud sites, 24 polar nodes; refine separately. Error bands exclude quadrature
error.
[WARNING] remaining model assumptions: Resolved spin/emission channels, diagonal Stark shifts, frozen isolated wells, no recapture;
Raman coupling is unsupported.
[WARNING] thermal observables: Energy and temperature describe survivors and include loss selection. Temperature is an energy-
matched proxy; coherent loss-selection decomposition is not recorded.
```

Bands / error bars: pointwise approximate 95% Monte Carlo intervals. Heatmaps show means.